Cheatsheet AZ-104 — 13 · Network security: NSG and ASG, effective rules, Bastion, service endpoints and private endpoints

1. Network Security Groups

Rule property	Values
Priority	**100-4096**; evaluated from **lowest to highest** and the first match **stops** the evaluation; no two rules can have the same priority and direction
Source / Destination	Any, IP or CIDR, **several IPs** (augmented), **service tag** (VirtualNetwork, Internet, AzureLoadBalancer, AzureCloud, Storage, Storage.WestEurope, Sql, AzureKeyVault, AzureBastion, GatewayManager, AzureMonitor...) or **application security group**
Protocol	TCP, UDP, ICMP, ESP, AH, Any
Port	Single, range or list (`80,443,8000-8010`)
Direction	Inbound / Outbound
Action	Allow / Deny

Default rule	Priority	What it does
AllowVnetInBound	65000	Allows everything coming from the **VirtualNetwork** tag (the VNet, **peered VNets** and **on-prem** via gateway)
AllowAzureLoadBalancerInBound	65001	Allows the load balancer **probes** (168.63.129.16)
DenyAllInBound	65500	Denies the rest (including **internet**)
AllowVnetOutBound	65000	Outbound to the VNet
AllowInternetOutBound	65001	Outbound to the **internet** allowed by default
DenyAllOutBound	65500	Denies the rest

- They cannot be deleted, but they are **overridden** by lower-numbered rules. For "deny all outbound internet" → Deny rule to Internet with priority < 65001 (and Allow to the required service tags).
- **Association:** one NSG to **subnets** and/or **NICs** (a subnet or NIC has only **one** NSG; one NSG serves many). It must be in the **same region** (and subscription) as the VNet. Limit **2,000 rules** per NSG.
- **Evaluation** with an NSG on the subnet **and** on the NIC: inbound → **subnet first, then NIC**; outbound → **NIC first, then subnet**. Traffic must be allowed in **both**. The subnet NSG also filters traffic **between VMs in the same subnet**.
- **Stateful:** an Inbound rule allows the replies; no Outbound rule is needed (and vice versa). Changing rules only affects **new connections**. LB probes and platform services (DHCP, DNS, IMDS at 168.63.129.16 / 169.254.169.254) are not blocked except with specific tags. Outbound port **25** blocked except on EA / MCA-E subscriptions (use an SMTP relay). **ICMP** is not allowed from the internet by default.

2. Application Security Groups

- They group **NICs** by role (web, app, db) and are used as **source or destination** in NSG rules: "Allow TCP 1433 from asg-web to asg-db". A NIC can belong to **several** ASGs; all NICs in an ASG must be in the **same VNet**, and if a rule uses two ASGs, both in the same VNet. They allow stable rules even when IPs change or VMs are added.

3. Interpreting effective rules (what shows up with tables)

Scenario	Result	Why
NIC NSG allows RDP from the internet; subnet NSG has no rule	**Blocked**	DenyAllInBound wins on the subnet; both are needed
Subnet NSG allows RDP; the NIC has no NSG	Allowed	Only the subnet one is evaluated
Rule 100 Allow 3389 from Any; rule 110 Deny 3389 from Any	**Allowed**	The lower number wins
Rule 100 Deny 3389 from Internet; rule 200 Allow 3389 from 203.0.113.0/24	Blocked for everyone	100 matches first (Internet includes that IP): swap the priorities
VM1 and VM2 in different subnets of the same VNet, no custom rules	They see each other	AllowVnetInBound
LB backend marked unhealthy after a new NSG	AzureLoadBalancer / 168.63.129.16 was blocked	Add Allow from the AzureLoadBalancer tag to the probe port
Deny Any → Internet outbound rule with priority 100	VMs do not update and cannot reach Storage	Add Allow to the tags (Storage, AzureMonitor...) with a lower priority number, or service endpoints
VM in a peered VNet cannot get in	NSG with source limited to the local subnet	VirtualNetwork includes peered VNets; review custom rules

- **Effective security rules** (NIC → Networking) shows the sum of subnet + NIC; **IP flow verify** tells which rule allows or denies a specific flow; **NSG diagnostics** evaluates all NSGs in the path (cheatsheet 12). **Flow logs** (VNet flow logs; NSG flow logs retire in Sept. 2027) record accepted / denied per rule.

4. Azure Bastion

Aspect	Detail
What it is	RDP / SSH to VMs **through the portal (HTML5, port 443)** or with a native client, with no **public IP** on the VMs and without opening 3389 / 22 to the internet
Requirements	`AzureBastionSubnet` **/26 or larger** subnet (same VNet and RG), **Standard static** public IP; reaches VMs in the VNet and in **peered VNets** (one Bastion in the hub); the VM needs an NSG rule allowing 3389 / 22 from the Bastion subnet (the VirtualNetwork tag covers it)
NSG on AzureBastionSubnet	Inbound: 443 from **Internet**, 443 from **GatewayManager**, 443 from **AzureLoadBalancer**, 8080 and 5701 from VirtualNetwork · Outbound: 3389 and 22 to VirtualNetwork, 443 to **AzureCloud**, 8080 and 5701 to VirtualNetwork, 80 to Internet
Developer	Free, no dedicated subnet, one connection at a time, portal only; for dev/test
Basic	2 instances; portal; peered VNets; Kerberos
Standard	**Host scaling** (2-50 instances), **native client** (mstsc / ssh / `az network bastion rdp`), **file transfer**, **shareable links**, connection by **IP** (on-prem via VPN), custom ports
Premium	Plus **session recording** and **private-only** deployment (no public IP)

- "Users must connect with their own RDP client through Bastion" → Standard. "Record RDP sessions for audit" → Premium. "Bastion deployment fails on /27" → the subnet must be /26.

5. Service endpoints vs private endpoints

	Service endpoint	Private endpoint (Private Link)
What it does	Extends the **subnet identity** to the service; traffic leaves with a **private** source IP over the backbone toward the service's **public endpoint**	Creates a **NIC with a private IP** from your subnet that represents the resource (blob, sqlServer, vault, sites...); the service becomes reachable as if it were in the VNet
Configured on	The **subnet** (`Microsoft.Storage`, `Microsoft.Sql`, `Microsoft.KeyVault`, `Microsoft.Web`, `Microsoft.ServiceBus`, `Microsoft.EventHub`, `Microsoft.AzureCosmosDB`, `Microsoft.ContainerRegistry`...) + **VNet rule** in the resource firewall	The **resource** (sub-resource) + subnet where the NIC lives + **private DNS zone** `privatelink.<service>`
DNS	Unchanged (resolves to the public IP)	Must resolve to the **private IP**: `privatelink.*` zone linked to the VNets (and forwarding from on-prem with **DNS Private Resolver**)
From on-prem	**No** (on-prem traffic arrives with NAT public IPs → IP rule)	**Yes**, via VPN / ExpressRoute
From peered VNets	Each subnet enables its own endpoint	Yes (the peering reaches the private IP; link the DNS zone to that VNet)
Public endpoint	Still exists (restricted with the firewall)	Can be **disabled** entirely ("public network access disabled")
Cost	Free	Per hour and per GB
Granularity	Whole account / server; **endpoint policies** (Storage) to limit to specific accounts	A specific resource and sub-resource; automatic or manual approval
When	Quick and cheap for "only from my VNet"	"No public endpoint", "from on-premises", "private IP", "zero trust"

Service	Private zone
Blob / File / Queue / Table / DFS / Web (static)	`privatelink.blob.core.windows.net` · `privatelink.file.core.windows.net` · `privatelink.queue...` · `privatelink.table...` · `privatelink.dfs...` · `privatelink.web...`
Azure SQL Database	`privatelink.database.windows.net`
Key Vault	`privatelink.vaultcore.azure.net`
App Service	`privatelink.azurewebsites.net`
Container Registry	`privatelink.azurecr.io`

- When a service endpoint is enabled, existing connections are dropped and the public IP rules for that subnet stop working (it now arrives with a private IP). For **Azure SQL** the endpoint only works in the **same region**; Storage has `Microsoft.Storage.Global` for other regions.
- Private endpoint and NSG: the private endpoint **network policies** can be enabled on the subnet to apply NSG / UDR to traffic toward the endpoint NIC. To expose **your own** service behind a Standard ILB to other tenants → **Private Link service**.

6. Other layers

Service	Purpose	Versus NSG
Azure Firewall	Managed, stateful firewall centralized in the hub (`AzureFirewallSubnet` /26): **network** rules, **application rules by FQDN**, **DNAT**, threat intelligence, Premium with IDPS and TLS inspection; **Firewall Policy** and Firewall Manager	NSG filters by IP / port on subnet or NIC; Firewall inspects and centralizes (all traffic arrives via **UDR**)
DDoS Protection	Network Protection (whole VNet) or IP Protection (per public IP); automatic mitigation	—
Just-in-time VM access (Defender for Cloud)	Opens 3389 / 22 **only on request** and for a limited time, creating a temporary NSG rule	"reduce exposure of management ports"

Service	Purpose	Versus NSG
Virtual Network Manager: security admin rules	Global rules evaluated **before** NSGs (Allow, Always allow, Deny)	Central governance
Web Application Firewall	On Application Gateway / Front Door (layer 7, OWASP)	Cheatsheet 14

7. Commands that show up

Task	PowerShell (Az.Network)	Azure CLI
Create NSG and rule	`$r = New-AzNetworkSecurityRuleConfig -Name AllowRDP -Priority 100 -Direction Inbound -Access Allow -Protocol Tcp -SourceAddressPrefix 203.0.113.0/24 -SourcePortRange * -DestinationAddressPrefix * -DestinationPortRange 3389; New-AzNetworkSecurityGroup -Name nsg1 -ResourceGroupName RG1 -Location westeurope -SecurityRules $r`	`az network nsg create -g RG1 -n nsg1` · `az network nsg rule create -g RG1 --nsg-name nsg1 -n AllowRDP --priority 100 --direction Inbound --access Allow --protocol Tcp --source-address-prefixes 203.0.113.0/24 --destination-port-ranges 3389`
Rule with tag or ASG	`-SourceAddressPrefix AzureLoadBalancer` · `-SourceApplicationSecurityGroupId <asg-web> -DestinationApplicationSecurityGroupId <asg-db>`	`--source-address-prefixes AzureLoadBalancer` · `--source-asgs asg-web --destination-asgs asg-db`
Modify / list rules	`$nsg = Get-AzNetworkSecurityGroup -Name nsg1 -ResourceGroupName RG1; Set-AzNetworkSecurityRuleConfig -NetworkSecurityGroup $nsg -Name AllowRDP -Access Deny ...; Set-AzNetworkSecurityGroup -NetworkSecurityGroup $nsg` · `Get-AzNetworkSecurityRuleConfig -DefaultRules`	`az network nsg rule update -g RG1 --nsg-name nsg1 -n AllowRDP --access Deny` · `az network nsg rule list -g RG1 --nsg-name nsg1 --include-default -o table`
Associate	Subnet: `Set-AzVirtualNetworkSubnetConfig ... -NetworkSecurityGroup $nsg; Set-AzVirtualNetwork` · NIC: `$nic.NetworkSecurityGroup = $nsg; Set-AzNetworkInterface`	`az network vnet subnet update -g RG1 --vnet-name vnet1 -n sub1 --network-security-group nsg1` · `az network nic update -g RG1 -n nic1 --network-security-group nsg1` · remove: `--network-security-group ""`
ASG	`New-AzApplicationSecurityGroup -Name asg-web -ResourceGroupName RG1 -Location westeurope` · on the NIC: `$nic.IpConfigurations[0].ApplicationSecurityGroups = @($asg); Set-AzNetworkInterface`	`az network asg create -g RG1 -n asg-web` · `az network nic ip-config update -g RG1 --nic-name nic1 -n ipconfig1 --application-security-groups asg-web`
Effective rules	`Get-AzEffectiveNetworkSecurityGroup -NetworkInterfaceName nic1 -ResourceGroupName RG1`	`az network nic list-effective-nsg -g RG1 -n nic1`
Bastion	`New-AzBastion -ResourceGroupName RG1 -Name bas1 -PublicIpAddressRgName RG1 -PublicIpAddressName pip-bas -VirtualNetworkRgName RG1 -VirtualNetworkName vnet1 -Sku Standard -ScaleUnit 4`	`az network vnet subnet create -g RG1 --vnet-name vnet1 -n AzureBastionSubnet --address-prefixes 10.0.255.0/26` · `az network bastion create -g RG1 -n bas1 --vnet-name vnet1 --public-ip-address pip-bas --sku Standard --scale-units 4 --enable-tunneling true`
Service endpoint	`Set-AzVirtualNetworkSubnetConfig ... -ServiceEndpoint Microsoft.Storage,Microsoft.Sql; Set-AzVirtualNetwork` + `Add-AzStorageAccountNetworkRule -VirtualNetworkResourceId <subnet id>`	`az network vnet subnet update -g RG1 --vnet-name vnet1 -n sub1 --service-endpoints Microsoft.Storage Microsoft.Sql` + `az storage account network-rule add -n st1 --vnet-name vnet1 --subnet sub1`
Private endpoint	`New-AzPrivateEndpoint -Name pe-st1 -ResourceGroupName RG1 -Location westeurope -Subnet $subnet -PrivateLinkServiceConnection (New-AzPrivateLinkServiceConnection -Name c1 -PrivateLinkServiceId <st id> -GroupId blob)` + `New-AzPrivateDnsZone -Name privatelink.blob.core.windows.net` + `New-AzPrivateDnsVirtualNetworkLink` + `New-AzPrivateDnsZoneGroup`	`az network private-endpoint create -g RG1 -n pe-st1 --vnet-name vnet1 --subnet sub1 --private-connection-resource-id <st id> --group-id blob --connection-name c1` · `az network private-dns zone create -g RG1 -n privatelink.blob.core.windows.net` · `az network private-dns link vnet create -g RG1 -z privatelink.blob.core.windows.net -n link1 -v vnet1 -e false` · `az network private-endpoint dns-zone-group create -g RG1 --endpoint-name pe-st1 -n zg --private-dns-zone privatelink.blob.core.windows.net --zone-name blob`

8. Decision tree

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What does the wording ask for?
├─ Filter by IP / port on a subnet or NIC ..... NSG (remember subnet and NIC must both allow)
├─ Rules by server role without IPs ..... Application security groups
├─ Find out which rule blocks a flow ..... IP flow verify / NSG diagnostics
├─ RDP / SSH without public IP or VPN ..... Azure Bastion (native client, files → Standard)
├─ Open RDP only on request and for a while ..... Just-in-time VM access
├─ PaaS reachable only from my VNet, free, quick ..... Service endpoint + VNet rule
├─ PaaS with no public endpoint, access from on-prem ..... Private endpoint + privatelink zone + DNS
├─ PaaS reachable from a peered VNet by private IP ..... Private endpoint + link the zone to that VNet
├─ FQDN filtering, inspection, centralized DNAT ..... Azure Firewall (hub + UDR)
├─ Protect public IPs from DDoS ..... DDoS Network / IP Protection
├─ Expose my service to another tenant by private IP ..... Private Link service (Standard ILB)
└─ See denied traffic ..... VNet flow logs + Traffic analytics
```

9. Signals → answer

Wording	Answer
"RDP allowed on the NIC NSG but still blocked"	The subnet NSG denies it: allow on both
"rule 100 allows, rule 200 denies"	100 wins (lower number)
"allow web tier to reach database tier on 1433, servers change often"	ASGs + NSG rule

Wording	Answer
"load balancer backend unhealthy after adding NSG"	Allow AzureLoadBalancer (168.63.129.16) to the probe port
"block all outbound internet but allow Azure Storage"	Deny Internet + Allow service tag Storage with a lower priority number (or service endpoint)
"VMs in peered VNet blocked"	VirtualNetwork covers the peering; review custom rules
"NSG can be applied to a VNet"	False: subnet or NIC only
"RDP without public IP from the portal"	Bastion
"Bastion needs native RDP client / file copy / shareable link"	Standard
"session recording"	Premium
"AzureBastionSubnet /27 error"	Minimum /26
"storage only from subnet, no cost"	Service endpoint Microsoft.Storage + VNet rule
"storage not reachable from on-premises after service endpoint"	Expected: add the on-prem NAT public IP or use a private endpoint
"SQL private endpoint resolves to public IP"	Missing `privatelink.database.windows.net` zone linked to the VNet (or custom DNS without forwarding)
"private endpoint from on-premises"	VPN / ExpressRoute + DNS Private Resolver inbound (or forwarder)
"disable public access to the storage account"	Public network access Disabled + private endpoint
"inspect outbound traffic by FQDN"	Azure Firewall (application rules)
"management ports exposed", "reduce attack surface"	JIT VM access

10. Mistakes to avoid

1. Forgetting that between the subnet NSG and the NIC NSG, a deny from either one wins.
2. Giving the Deny rule a lower number than the specific Allow (or the reverse) without thinking about the order.
3. Blocking 168.63.129.16 / AzureLoadBalancer and breaking probes, DHCP or DNS.
4. Associating an NSG to a VNet (not possible) or from another region.
5. Creating the Bastion subnet with another name or with /27.
6. Expecting a service endpoint to work from on-prem or to change DNS.
7. Creating the private endpoint without the private DNS zone linked to every VNet that uses it.
8. Choosing NSG when the ask is FQDN filtering or centralized inspection (Azure Firewall).